

Chapter 4 Exam Review

Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Approximate the number using a calculator. Round your answer to three decimal places.

1) $3^{3.6}$ 1) _____

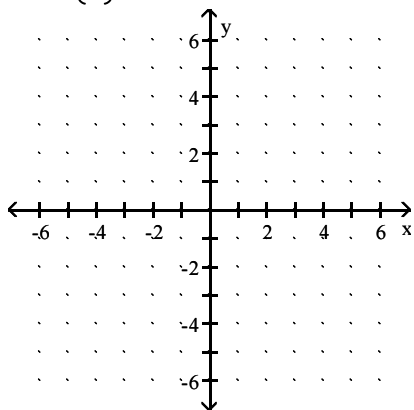
2) $5\sqrt{6}$ 2) _____

Solve the problem.

- 3) The rabbit population in a forest area grows at the rate of 8% monthly. If there are 270 rabbits in September, find how many rabbits (rounded to the nearest whole number) should be expected by next September. Use $y = 270(2.7)^{0.08t}$. 3) _____

Graph the function by making a table of coordinates.

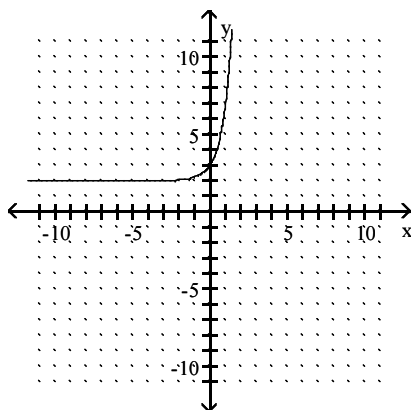
4) $f(x) = \left(\frac{1}{5}\right)^x$ 4) _____



MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

The graph of an exponential function is given. Select the function for the graph from the functions listed.

5) 5) _____



A) $f(x) = 5^x$

B) $f(x) = 5^x - 2$

C) $f(x) = 5^x + 2$

D) $f(x) = 5^x + 2$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Use the compound interest formulas $A = P\left(1 + \frac{r}{n}\right)^{nt}$ and $A = Pe^{rt}$ to solve.

- 6) Find the accumulated value of an investment of \$1000 at 6% compounded semiannually for 9 years. 6) _____

Write the equation in its equivalent exponential form.

7) $\log_5 x = 2$ 7) _____

8) $\log_b 243 = 5$ 8) _____

Write the equation in its equivalent logarithmic form.

9) $7^2 = 49$ 9) _____

10) $3^{-2} = \frac{1}{9}$ 10) _____

11) $\sqrt[3]{125} = 5$ 11) _____

Evaluate the expression without using a calculator.

12) $\log_5 25$ 12) _____

13) $\log_{12} 1$ 13) _____

14) $\log_8 8$ 14) _____

15) $\log_5 5^{17}$ 15) _____

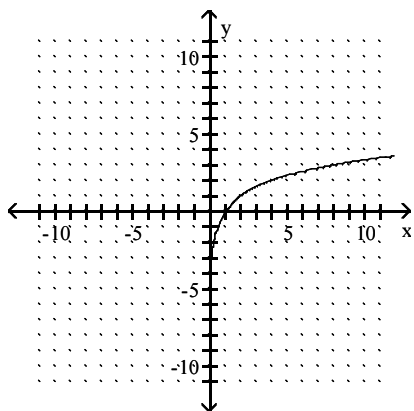
16) $2^{\log_2 10}$ 16) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

The graph of a logarithmic function is given. Select the function for the graph from the options.

17)

17) _____



A) $f(x) = \log_2 (x + 2)$

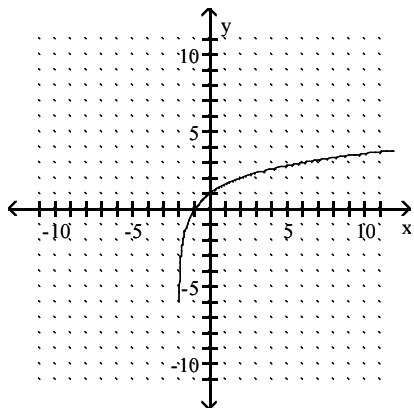
B) $f(x) = \log_2 x + 2$

C) $f(x) = \log_2 x$

D) $f(x) = \log_2 (x - 2)$

18)

18) _____



A) $f(x) = \log_2 x + 2$

B) $f(x) = \log_2 (x + 2)$

C) $f(x) = \log_2 x$

D) $f(x) = \log_2 (x - 2)$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the domain of the logarithmic function.

19) $f(x) = \log_9 (x + 7)$

19) _____

20) $f(x) = \ln (5 - x)$

20) _____

Use properties of logarithms to expand the logarithmic expression as much as possible. Where possible, evaluate logarithmic expressions without using a calculator.

21) $\log_5 (2 \cdot 11)$

21) _____

22) $\log_2 (4x)$

22) _____

23) $\log_3 \left(\frac{7}{13} \right)$

23) _____

24) $\ln \left(\frac{e^5}{6} \right)$

24) _____

25) $\log_4 11^{-2}$

25) _____

26) $\ln \sqrt[3]{x}$

26) _____

Use properties of logarithms to condense the logarithmic expression. Write the expression as a single logarithm whose coefficient is 1. Where possible, evaluate logarithmic expressions.

27) $3 \log_y 4 + \log_y 3$

27) _____

28) $\log_3 135 - \log_3 5$

28) _____

29) $\ln x + 8 \ln y$

29) _____

30) $\frac{1}{2} \log_9 x + \log_9 y$

30) _____

31) $\ln x + 3 \ln y$

31) _____

Answer Key

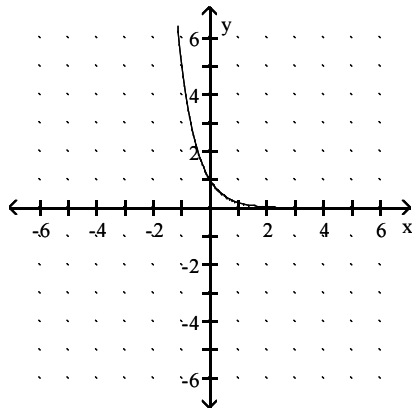
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1) 52.196

2) 51.537

3) 701

4)



5) D

6) \$1702.43

7) $5^2 = x$

8) $b^5 = 243$

9) $\log_7 49 = 2$

10) $\log_3 \frac{1}{9} = -2$

11) $\log_{125} 5 = \frac{1}{3}$

12) 2

13) 0

14) 1

15) 17

16) 10

17) C

18) B

19) $(-7, \infty)$

20) $(-\infty, 5)$

21) $\log_5 2 + \log_5 11$

22) $2 + \log_2 x$

23) $\log_3 7 - \log_3 13$

24) $5 - \ln 6$

25) $-2 \log_4 11$

26) $\frac{1}{3} \ln x$

27) $\log_y 192$

28) 3

29) $\ln xy^8$

30) $\log_9 y \sqrt{x}$

31) $\ln xy^3$